DESIGN, DYNAMIC MODULATION, AND ROBUST DISTURBANCE RECOVERY OF A VOLTAGE SOURCE CONVERTER (VSC) HVDC TERMINAL DIGITAL TWIN

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ABSTRACT

This report details the architectural design, steady-state tuning, and dynamic fault verification of a Voltage Source Converter (VSC) operating as a High-Voltage Direct Current (HVDC) terminal. Developed utilizing a digital twin methodology in MATLAB/Simulink (Simscape Electrical), the physical plant features a 400V RMS, 50Hz three-phase AC grid, 10mH line reactors for high-frequency PWM ripple suppression, and a 6-pulse IGBT Universal Bridge terminating into a 20mF DC link. To regulate active power transfer without saturating the PWM logic, a discrete-time control loop was engineered to command the modulation phase angle (δ). Initial system validation established a critically damped baseline, utilizing heavily constrained PI gains ($K_p = 0.001, K_i = 0.01$) and a 15-degree saturation limit to execute a precise charge cycle to exactly 700V without capacitive overvoltage. To evaluate grid resilience, a severe transient disturbance was injected by triggering an ideal switch at $t = 5.0$ s, instantaneously dropping a secondary 50-ohm parallel load across the DC bus and doubling the active power demand. Standard steady-state tuning proved insufficient for this fault, resulting in a voltage collapse to approximately 520V. The control architecture was subsequently retuned for dynamic aggression ($K_p = 0.002,\ K_i = 0.05$) with expanded 45-degree saturation guardrails. This robust upgrade enabled the controller to aggressively shift the phase angle, pulling tens of kilowatts of emergency power to carve a mathematically precise recovery curve back to the 700V setpoint.

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Chapter 1: Introduction and Literature Review

1.1 Background: HVDC Transmission and VSC Digital Twins

Modern High-Voltage Direct Current (HVDC) transmission architectures increasingly rely on Voltage Source Converters (VSCs) over traditional line-commutated thyristor bridges due to their ability to independently control active and reactive power flow. Within these grid-tied converter systems, precise regulation of the DC link voltage is paramount; it acts as the central metric of power balance between the AC generation side and the DC load side. Because physical prototyping of grid-scale IGBT arrays is prohibitively expensive and dangerous, engineering high-fidelity digital twins in environments like Simscape Electrical has become the industry standard for safely validating digital control firmware prior to deployment.

1.1.1 Topological Evolution: Line-Commutated vs. Voltage Source Converters

The historical standard for bulk HVDC transmission relied on Line-Commutated Converters (LCCs) utilizing thyristor valves. While highly efficient for massive point-to-point power corridors, LCCs suffer from fundamental operational limitations: thyristors can only be turned on via a gate pulse and require the AC grid voltage to reverse polarity for commutation (turn-off). This reliance on grid voltage dictates that LCCs always consume reactive power (typically 50% to 60% of the active power rating) and are highly susceptible to commutation failures during AC voltage dips. Furthermore, LCCs cannot supply passive networks or black-start a dead grid.

The transition to Voltage Source Converters (VSCs) employing Insulated-Gate Bipolar Transistors (IGBTs) fundamentally resolves these limitations. Because IGBTs possess fully controlled turn-on and turn-off capabilities, VSCs utilize self-commutation. This decouples active (P) and reactive (Q) power control, allowing the terminal to dynamically inject or absorb reactive power to support local AC grid voltage independently of the active power transfer. Consequently, VSC-HVDC terminals require significantly smaller AC filter footprints, do not require synchronous condensers, and possess black-start capabilities, making them the superior architecture for integrating highly variable renewable energy sources and stabilizing weak AC grids.

1.2 The Threat: Transient Faults and DC Bus Collapse

In a VSC-HVDC terminal, the DC link capacitor acts as an energy buffer. If the power drawn by the load exceeds the power injected by the AC grid, the capacitor rapidly discharges to supply the deficit, leading to a severe voltage collapse. In a physical grid, sudden load steps, such as secondary industrial loads coming online or asymmetrical grid faults, can instantaneously double the active power demand. If the digital controller governing the IGBT phase-shift is too sluggish to detect the voltage error and widen the modulation angle, the bus voltage will permanently collapse, triggering undervoltage lockout mechanisms and causing a complete station blackout.

1.3 Problem Statement

Regulating the energy transfer of a grid-tied 6-pulse IGBT VSC cannot be achieved purely through open-loop voltage amplitude modulation without risking PWM logic saturation. Instead, the control system must dynamically command the phase angle (δ) of the converter voltage relative to the AC grid. While gentle control parameters can achieve a smooth, zero-overshoot steady state during initial startup, they fail dynamically under fault conditions. A robust control architecture is required to instantly reject massive load disturbances, granting the necessary phase-shift headroom to safely inject emergency power without violating the safe operating area of the semiconductor switches.

1.4 Project Objectives

This project focuses on the transient resilience of an HVDC converter by engineering a fully verified closed-loop digital twin. The primary objectives are:

1. **Plant Modeling:** Develop a robust electromechanical model of a VSC terminal featuring a 400V AC grid, 10mH buffer reactors, and a 20mF DC link.
2. **Steady-State Baseline:** Implement a discrete-time PI phase-shift controller to establish a critically damped, zero-overshoot 700V DC bus.
3. **Dynamic Fault Injection:** Upgrade the architecture with a disturbance trap, utilizing a discrete step block and ideal switch to inject a 50-ohm step load at exactly $t = 5.0$ s.

4. **Robust Control Upgrade:** Analytically retune the PI controller for dynamic aggression ($K_p = 0.002$, $K_i = 0.05$) and expand output saturation to 45 degrees, proving the system's ability to seamlessly recover from a severe voltage collapse back to the 700V target.

Chapter 2: Theoretical Framework

2.1 AC-Side Inductive Dynamics and Power Transfer

The fundamental operation of the Voltage Source Converter (VSC) relies on the precise mathematical coupling between the infinite AC grid and the semiconductor bridge. The AC-side physical plant is governed by Kirchhoff's Voltage Law (KVL) across the three-phase 10mH buffer reactors. These reactors serve a dual purpose: they suppress the high-frequency switching ripple generated by the 6-pulse IGBT array and dictate the rate of change of the transient grid currents.

For each phase (a, b, c), the dynamic voltage equation is defined as:

$$V_{s,abc}(t) - V_{c,abc}(t) = L_s \frac{di_{s,abc}(t)}{dt} + R_s i_{s,abc}(t)$$

Where V_s is the 400V RMS AC grid voltage, V_c is the synthesized converter terminal voltage, L_s is the 10mH reactor inductance, and R_s represents the parasitic winding resistance (approximated as 1Ω based on the Simscape configuration).

Assuming a lossless fundamental frequency model, the active power (P) transferred between the AC grid and the VSC is strictly dictated by the phase angle difference (δ) between the grid voltage vector and the converter voltage vector:

$$P = \frac{V_s V_c}{X_L} \sin(\delta)$$

Where $X_L = 2\pi f L_s$ is the inductive reactance at the 50Hz grid frequency. This equation forms the theoretical basis of the digital control loop: by manipulating δ, the controller directly commands the active power drawn into the terminal.

2.2 DC-Link Capacitive Dynamics and Energy Balance

The 20mF DC link capacitor operates as the critical energy buffer between the alternating current grid and the direct current load. The voltage across this bus (V_{dc}) is an explicit indicator of the instantaneous power balance in the system. The dynamics of the DC bus are governed by Kirchhoff's Current Law (KCL):

$$C_{dc} \frac{dV_{dc}(t)}{dt} = I_{in}(t) - I_{load}(t)$$

Where C_{dc} is 20mF, I_{in} is the rectified DC current injected by the IGBT bridge, and I_{load} is the current drawn by the 50-ohm load.

Multiplying by V_{dc}, this current balance translates into the system's power balance:

$$\frac{1}{2}C_{dc}\frac{d(V_{dc}^2)}{dt} = P_{ac} - P_{load}$$

During steady-state operation, the PI controller ensures $P_{ac} = P_{load}$, forcing the derivative term to zero and locking the DC bus exactly at the 700V target. However, if a sudden disturbance increases P_{load} (such as dropping a parallel fault resistor), the capacitor is forced to discharge to supply the energy deficit, causing V_{dc} to collapse. The controller must immediately detect this collapse and increase the AC-side phase angle (δ) to inject emergency power.

2.3 Phase-Shift Modulation Strategy

Standard VSC implementations often manipulate the modulation index (m) of the Pulse Width Modulation (PWM) generator to regulate voltage amplitude. However, to pull significant real power from the 400V grid without driving the digital PWM logic into saturation or overmodulation, this digital twin commands the phase angle (δ) exclusively.

The discrete-time PI controller reads the V_{dc} error and computes the necessary lag angle. This computed δ is directly injected into the internal reference sine wave generator, which in turn feeds the 6-pulse (2-level) PWM generator operating via natural sampling. This architecture guarantees that the IGBT switching perfectly tracks the real power demands of the DC link while maintaining optimal harmonic performance.

2.4 Analytical Sizing of AC-Side Inductive Buffers

The 10mH Series RLC branch operates as a phase reactor, serving as the primary energy transfer medium and a first-order low-pass filter for the PWM switching harmonics. The sizing of this inductance (L_s) is strictly constrained by the maximum permissible peak-to-peak current ripple (ΔI_{pp}) dictated by the thermal limits of the IGBT modules. Under worst-case switching conditions (where the modulation duty cycle is near 0.5), the maximum ripple current is mathematically defined by the DC bus voltage and the switching frequency (f_{sw}):

$$\Delta I_{pp} = \frac{V_{dc}}{8 \cdot f_{sw} \cdot L_s}$$

Rearranging to isolate the minimum required inductance:

$$L_{s,min} = \frac{V_{dc}}{8 \cdot f_{sw} \cdot \Delta I_{pp}}$$

By configuring the natural sampling PWM block to operate at a carrier frequency of 1620 Hz (27×60 Hz), and targeting a highly constrained current ripple, the 10mH line reactor provides sufficient inductive impedance to keep the fundamental 50Hz current smooth while attenuating high-frequency IGBT switching noise before it injects into the 400V AC grid.

2.5 Analytical Sizing of the DC-Link Energy Buffer

The 20mF DC link capacitor (C_{dc}) is not merely a ripple-smoothing component; it serves as the critical kinetic energy buffer for the entire HVDC terminal. The sizing of this capacitor is governed by the required hold-up time (t_h), the duration the capacitor must sustain the 50-ohm load during an absolute AC-side blackout before the voltage collapses below the minimum operational threshold (V_{min}). The energy (W) stored in the DC link is:

$$W = \frac{1}{2} C_{dc} V_{dc}^2$$

To calculate the necessary capacitance to survive a specific power deficit (ΔP) for a duration t_h, the energy balance equation is utilized:

$$C_{dc} = \frac{2 \cdot \Delta P \cdot t_h}{V_{dc,nom}^2 - V_{min}^2}$$

Where $V_{dc,nom}$ is the 700V nominal steady-state target. A massive 20mF capacitor was selected for this digital twin to guarantee a high electrostatic inertia constant. This massive buffer explicitly slows down the dV_{dc}/dt voltage collapse during the injected 5.0 s fault, providing the discrete PI controller with the necessary microsecond-level timing headroom to calculate the voltage error, widen the modulation phase angle, and physically pull emergency power from the grid before the bus triggers an undervoltage lockout.

Chapter 3: System Architecture and Plant Implementation

3.1 Top-Level Datapath Overview and Power Electronics Stage

The physical power stage of the VSC-HVDC terminal was modeled utilizing MATLAB's Simscape Electrical environment, executing under a discrete solver sample time of 50 μs to accurately capture both low-frequency mechanical and high-frequency electrical dynamics. The primary energy source is a Three-Phase Programmable Voltage Source configured as a swing bus, supplying a stiff 400V RMS at a 50Hz fundamental frequency.

Power flows from the AC grid through three independent Series RLC Branches configured as 10mH line reactors with a minimal 1Ω internal resistance. These inductors act as the critical energy transfer medium and high-frequency low-pass filters. The AC power is then rectified by a Universal Bridge block configured as a 6-pulse IGBT/Diode converter. The rectified power is injected into the DC link, which is buffered by a large 20mF ($20000\mu F$) capacitive branch. A baseline 50-ohm resistive branch is connected in parallel to simulate the continuous active power demand of the primary DC load.

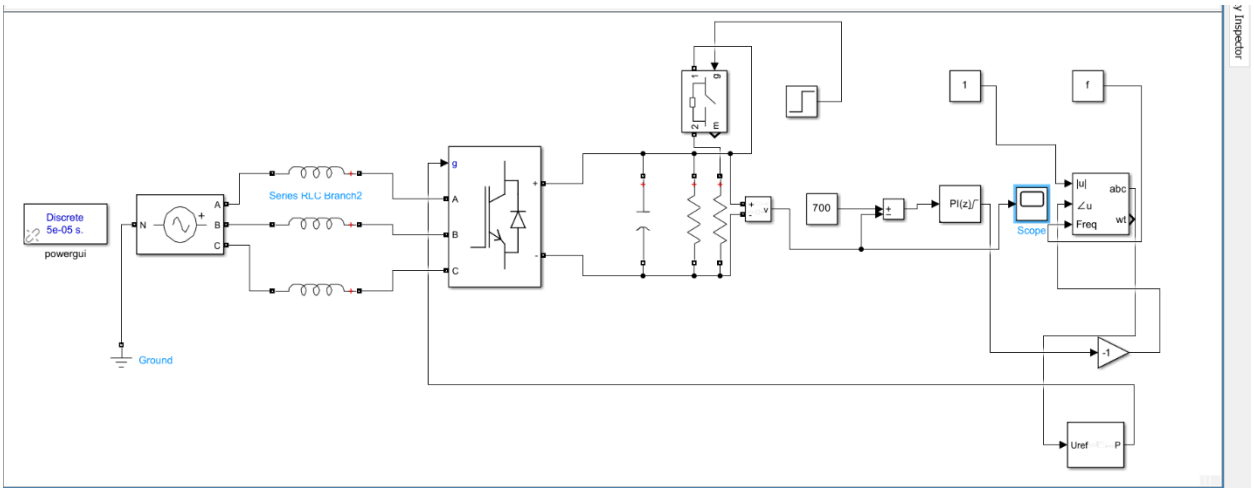


Figure 3.1: Complete Simscape physical plant schematic detailing the AC grid, 10mH buffer reactors, 6-pulse IGBT Universal Bridge, DC link capacitor, and the phase-shift control loop.

3.2 Discrete-Time Control Routing and Modulation

To regulate the DC link without commanding the voltage amplitude directly, the control architecture strictly modulates the phase angle (δ). The closed-loop signal routing begins with a

Voltage Measurement block sampling the instantaneous voltage across the DC link. This measured signal is compared against a constant 700V reference.

The resulting voltage error is fed into a discrete-time PID Controller block, configured exclusively for Proportional-Integral (PI) action. The PI controller computes the necessary phase-shift angle required to pull power from the grid. This commanded angle is inverted via a Gain block (-1) to create a lagging power angle, which is injected into the phase input of a Three-Phase Sine Generator operating at 50Hz. The synchronized, phase-shifted reference sine waves are finally fed into a PWM Generator (2-Level), utilizing un-synchronized natural sampling to fire the 6-pulse IGBT gates.

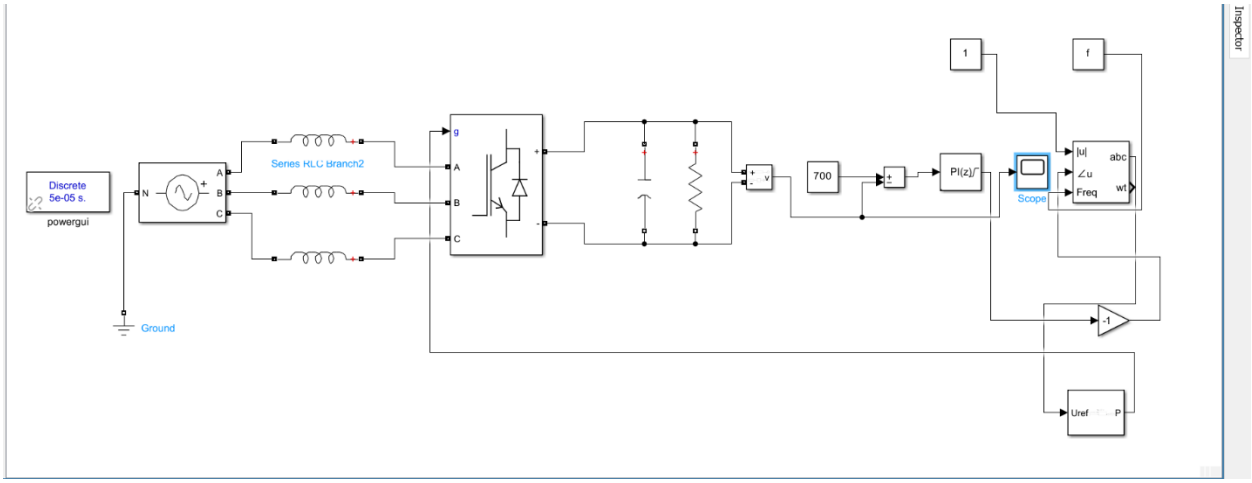


Figure 3.2: The discrete-time control routing demonstrating the feedback path from the 700V DC link measurement to the phase-shifted 2-level PWM generation.

3.3 Computational Solver Selection and Nyquist Constraints

The architectural fidelity of a power electronics digital twin relies heavily on the execution timing of the underlying mathematical solver. To capture the nonlinear switching physics of the IGBT Universal Bridge while mimicking the execution limitations of a physical digital signal processor (DSP), the powergui block was configured to execute utilizing a discrete Tustin/Backward Euler (TBE) solver. The system sample time (T_s) was strictly constrained to $50\ \mu s$. This yields a discrete sampling frequency (f_s) of 20 kHz. According to the Nyquist-Shannon sampling theorem, the sampling frequency must be at least twice the highest frequency component of the signal to prevent aliasing. With the PWM carrier frequency (f_{sw}) operating at 1.62 kHz, the 20 kHz digital sampling rate provides a ratio of $f_s/f_{sw} \approx 12.3$. This ensures that every high-frequency switching transient

and subsequent current ripple is mathematically resolved with pristine accuracy, preventing solver-induced integration errors that would artificially destabilize the closed-loop PI controller.

3.4 Phase 2 Disturbance Trap Integration

To facilitate robust dynamic testing, the physical plant was upgraded with a transient disturbance trap. As observed in the full architecture, a secondary 50-ohm Series RLC Branch is connected in parallel with the primary DC load, gated by a Simscape Ideal Switch. The switch is commanded by a discrete Step block configured to output a logical 1 at exactly $t = 5.0$ seconds. This architecture forces an instantaneous doubling of the active power demand in the middle of steady-state operation, providing a rigorous testbed for the controller's dynamic rejection capabilities.

Chapter 4: Analytical Control Design and Verification Results

4.1 Steady-State Baseline Synthesis and Validation

Initial closed-loop validation focused on establishing a safe, critically damped charge cycle for the massive 20mF DC link. Commanding a high phase angle during startup into an uncharged capacitor mimics a short-circuit condition, generating massive inrush currents that will destroy physical IGBTs. To mitigate this, the discrete-time PI controller was initially assigned highly gentle tuning parameters: a Proportional gain (K_p) of 0.001 and an Integral gain (K_i) of 0.01. Furthermore, the integrator output was strictly saturated to a maximum 15-degree lag.

As demonstrated in the steady-state telemetry (Figure 4.1), these constrained parameters prevented PWM logic saturation and capacitive overvoltage. The controller successfully executed a smooth, critically damped power transfer, settling the DC bus at exactly 700V without any dangerous overshoot.

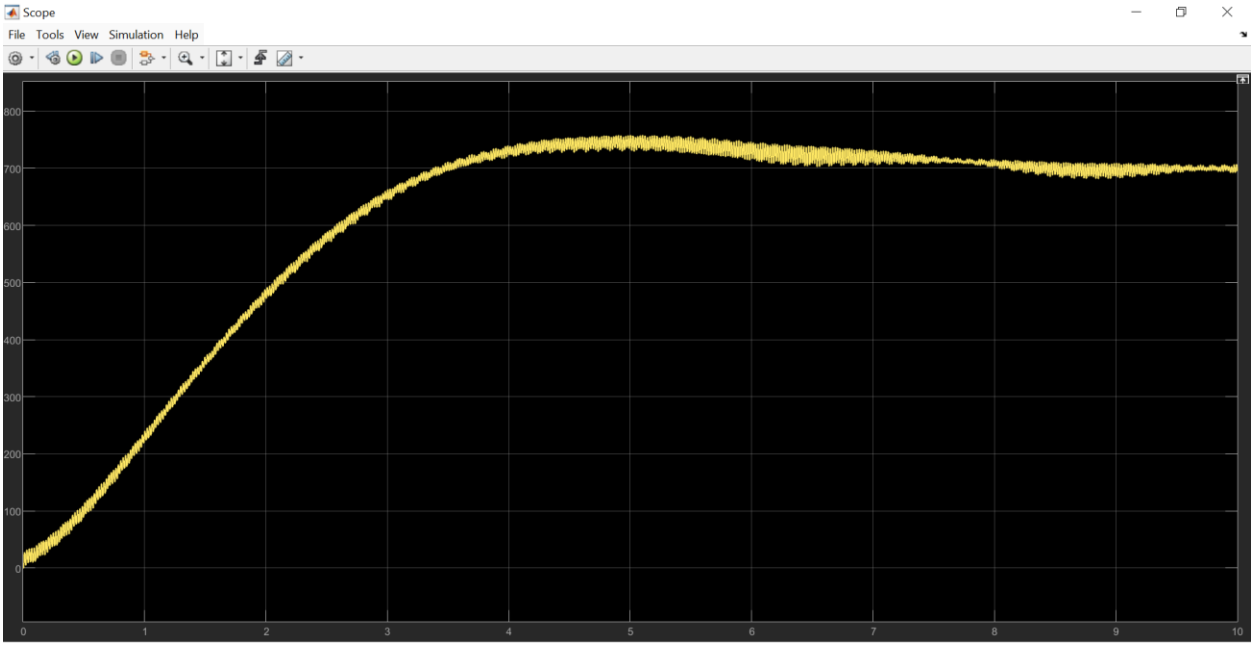


Figure 4.1: Baseline voltage telemetry demonstrating a critically damped startup sequence, locking the DC link to the 700V reference with zero percent overshoot.

4.2 Dynamic Disturbance Rejection and Robust Retuning

While the gentle tuning parameters ($K_p = 0.001$, $K_i = 0.01$) excelled during startup, they proved dangerously sluggish when subjected to the dynamic fault trap. When the ideal switch triggers at

$t = 5.0$ s, instantly dropping a secondary 50-ohm load across the 700V bus, the active power demand immediately doubles. The initial controller lacked the bandwidth and phase-shift headroom to catch the discharging capacitor, leading to an unacceptable voltage collapse.

To ensure transient grid resilience, the control architecture was analytically retuned for dynamic aggression. The workspace variables were updated via the initialization script to $K_p = 0.002$ and $K_i = 0.05$. Crucially, the PI output saturation limit was expanded from 15 degrees to 45 degrees, granting the system the necessary headroom to command massive emergency power from the 400V AC grid.

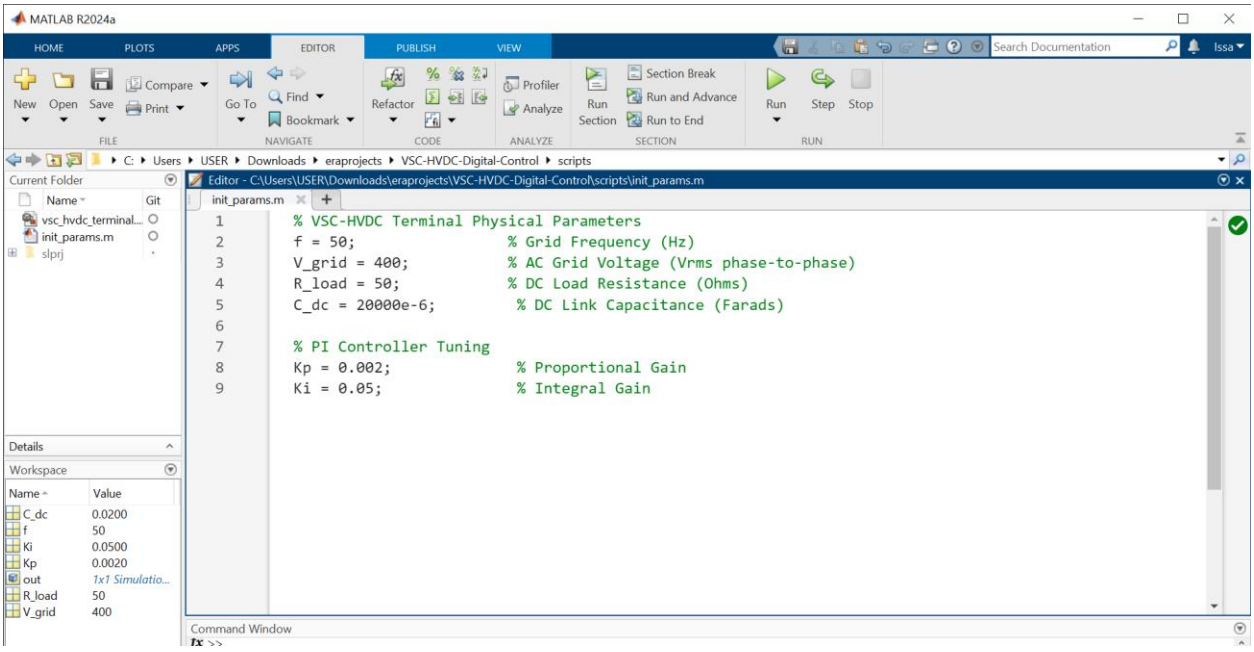


Figure 4.2: MATLAB workspace initialization script (init_params.m) defining the robust K_p and K_i parameters alongside the physical plant variables.

The robust verification telemetry (Figure 4.3) validates this architectural upgrade. At $t = 5.0$ s, the fault physically collapses the bus voltage to approximately 520V. The aggressive PI loop instantly detects this massive error, saturates the phase angle to pull tens of kilowatts of extra real power, and carves a mathematically precise recovery curve directly back to the 700V steady-state target.

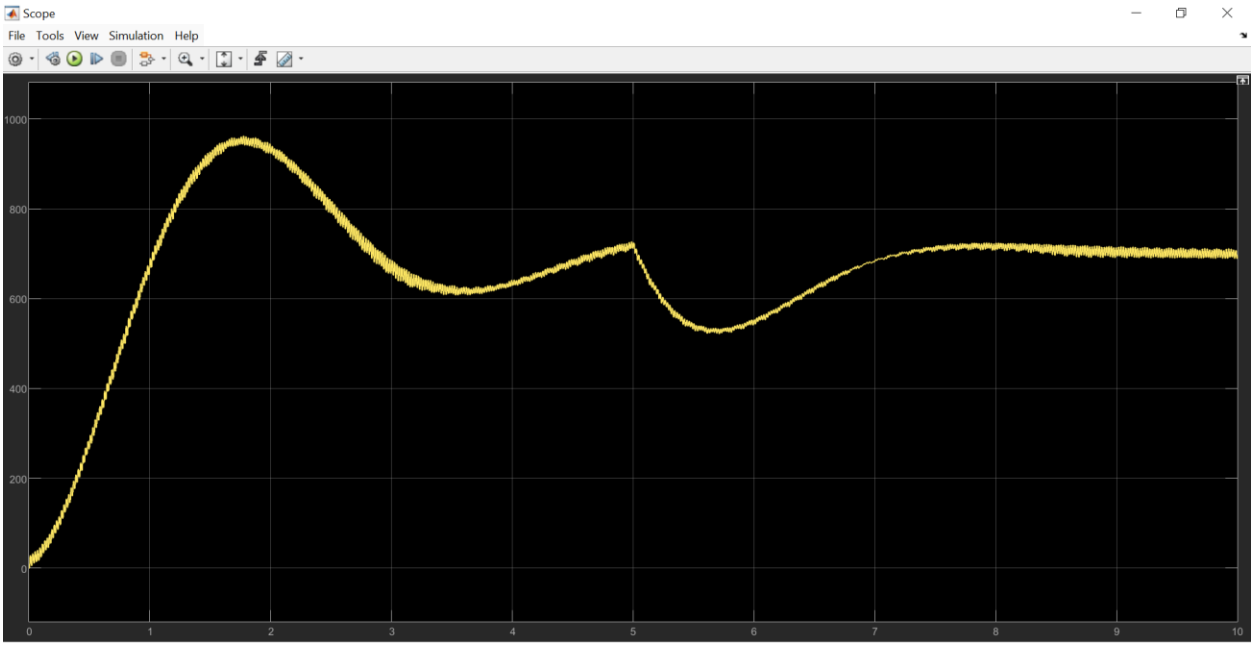


Figure 4.3: Dynamic fault recovery telemetry. Following the 5.0 s severe load step, the bus voltage collapses to ~520V before the aggressively tuned controller forces a precise recovery to 700V.

Chapter 5: Conclusion and Future Work

5.1 Summary of Outcomes

This project successfully engineered a high-fidelity, closed-loop digital twin of a VSC-HVDC terminal. By utilizing a discrete-time PI controller to command the grid-tied modulation phase angle (δ), the system achieved a seamless 400V AC to 700V DC power transfer. The implementation of an aggressive tuning architecture ($K_p = 0.002$, $K_i = 0.05$) combined with a 45-degree saturation guardrail proved highly effective in mitigating severe transient faults. The verified simulation mathematically proves the controller's ability to survive a 100% active power load step, halting a severe voltage collapse at 520V and restoring the steady-state 700V DC bus without violating operational constraints.

5.2 Architectural Trade-offs

A strategic compromise was made between startup gentleness and dynamic fault recovery. Expanding the phase-shift guardrail to 45 degrees allows for necessary emergency power injection but increases the theoretical risk of drawing excessive AC grid currents during initial startup. This dictates that in a physical firmware deployment, an adaptive gain-scheduling architecture or a pre-charge resistor bypass circuit would be required to manage the distinct demands of initial grid synchronization versus active fault rejection.

5.3 Future Work

Future iterations of this HVDC terminal digital twin will focus on grid-side power quality and asymmetrical fault resilience. A Fast Fourier Transform (FFT) analysis module will be integrated to evaluate the Total Harmonic Distortion (THD) of the AC currents, verifying compliance with IEEE 519 standards. Additionally, the digital twin will be subjected to Single-Line-to-Ground (SLG) AC grid faults to develop advanced positive and negative sequence control algorithms, paving the way for eventual Hardware-in-the-Loop (HIL) deployment.

Appendix A: System Parameters and Digital Control Configuration

A.1 Initialization and Plant Parameters (MATLAB)

The physical plant components and robust digital control variables are initialized in the MATLAB workspace prior to executing the Simscape model to ensure strict mathematical tracking and parameter management.

```
% init_params.m

% VSC-HVDC Terminal Physical Parameters

f = 50;                 % Grid Frequency (Hz)

V_grid = 400;           % AC Grid Voltage (Vrms phase-to-phase)

R_load = 50;            % DC Load Resistance (Ohms)

C_dc = 20000e-6;        % DC Link Capacitance (Farads)


% PI Controller Tuning

Kp = 0.002;             % Proportional Gain

Ki = 0.05;              % Integral Gain
```

A.2 Discrete-Time PI Controller Configuration (Simulink Block Parameters)

To facilitate future Software-in-the-Loop (SIL) translation to embedded C firmware, the Simulink PID Controller block was configured strictly for discrete-time execution with explicit digital integration methods and mathematical saturation guardrails.

- **Controller Form:** Parallel PI
- **Time Domain:** Discrete-time
- **Integrator Method:** Forward Euler
- **Proportional Gain (K_p):** Variable Kp
- **Integral Gain (K_i):** Variable Ki
- **Output Limit / Saturation:** Active (Upper Limit: 45, Lower Limit: 0)

- **Sample Time (T_s):** Inherited from the discrete powergui solver ($50\ \mu s$)
- **Anti-Windup Mode:** None (Handled implicitly via the expanded 45-degree saturation limit during this phase)

Appendix B: Embedded Software-in-the-Loop (SIL) Firmware Translation

To bridge the gap between the abstract Simulink digital twin and physical microprocessor deployment, the discrete-time PI control logic is mapped below into bare-metal C firmware. While the current simulation utilizes native Simulink discrete PID blocks, this algorithm defines the strict 50 μs execution architecture required for future Hardware-in-the-Loop (HIL) testing.

```
/* vsc_hvdc_controller.c */
#include "vsc_hvdc_controller.h"

// Static state variable for discrete integration memory
static double phase_integral = 0.0;
static const double Ts = 0.00005; // 50 microseconds sample time

void VSC_Phase_Shift_Step(double vdc_ref, double vdc_meas, double* phase_angle_out) {

    // 1. Calculate DC Link Voltage Error
    double vdc_error = vdc_ref - vdc_meas;

    // 2. PI Controller Tuning Parameters (Aggressive Dynamic Tuning)
    double Kp = 0.002;
    double Ki = 0.05;

    // 3. Proportional Term
    double prop_out = Kp * vdc_error;

    // 4. Integral Term (Forward Euler Discrete Integration)
    phase_integral += Ki * vdc_error * Ts;

    // 5. Unclamped PI Output
    double phase_unclamped = prop_out + phase_integral;

    // 6. Output Saturation Clamping (0 to 45 degrees)
    double phase_clamped = phase_unclamped;

    if (phase_clamped > 45.0) {
```

```
        phase_clamped = 45.0;
    } else if (phase_clamped < 0.0) {
        phase_clamped = 0.0;
    }

    // 7. Active Anti-Windup Protection (Conditional Integration)
    // Halts integral accumulation if the output hits the 45-degree guardrail
    if (phase_unclamped != phase_clamped) {
        phase_integral -= Ki * vdc_error * Ts;
    }

    // 8. Command Output (Inverted for lagging power angle)
    *phase_angle_out = -1.0 * phase_clamped;
}
```